

GEN211 Probability and Statistics

Lecture 6: Discrete Probability Distributions





Lecture contents

- Mean, variance, standard deviation, and expectation for discrete probability distributions
- The binomial, Poisson, and geometric distributions



Mean, Variance, Standard Deviation, and Expectation for Discrete Probability Distributions



Discrete probability distributions

A **random variable** is a variable whose values are determined by chance.

A **discrete probability distribution** consists of the values a random variable can assume and the corresponding probabilities of the values. The probabilities are determined theoretically or by observation.

Construct a probability distribution for rolling a single die.

Since the sample space is 1, 2, 3, 4, 5, 6 and each outcome has a probability of $\frac{1}{6}$, the distribution is as shown.

Outcome X	1	2	3	4	5	6
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$



Constructing discrete probability distributions

The procedure shown here for constructing a probability distribution for a discrete random variable uses the probability experiment of tossing three coins. Recall that when three coins are tossed, the sample space is represented as TTT, TTH, THT, HTT, HHT, HTH, THH, HHH; and if X is the random variable for the number of heads, then X assumes the value 0, 1, 2, or 3.

Probabilities for the values of X can be determined as follows:

No heads	One head			Two heads			Three heads
TTT	TTH	THT	HTT	HHT	HTH	THH	HHH
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{8}$	$\frac{3}{8}$			$\frac{3}{8}$			$\frac{1}{8}$

Hence, the probability of getting no heads is $\frac{1}{8}$, one head is $\frac{3}{8}$, two heads is $\frac{3}{8}$, and three heads is $\frac{1}{8}$. From these values, a probability distribution can be constructed by listing the outcomes and assigning the probability of each outcome, as shown here.

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

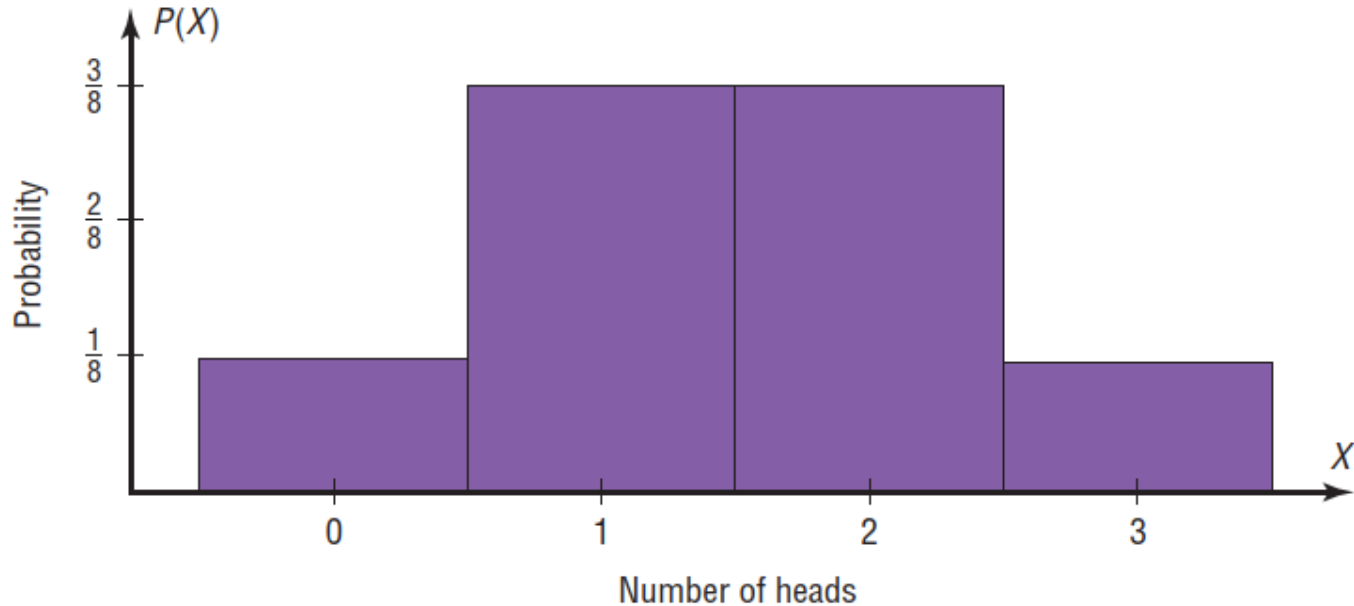


Constructing discrete probability distributions

Represent graphically the probability distribution for the sample space for tossing three coins.

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

The values that X assumes are located on the x axis, and the values for $P(X)$ are located on the y axis.



Example 1

The baseball World Series is played by the winner of the National League and that of the American League. The first team to win four games wins the World Series. In other words, the series will consist of four to seven games, depending on the individual victories. The data shown consist of 40 World Series events. The number of games played in each series is represented by the variable X . Find the probability $P(X)$ for each X , construct a probability distribution, and draw a graph for the data.

X	Number of series played
4	8
5	7
6	9
7	16
	Total 40



Example 1

SOLUTION

The probability $P(X)$ can be computed for each X by dividing the number of series played for each X by the total.

$$\text{For 4 games, } \frac{8}{40} = 0.200$$

$$\text{For 6 games, } \frac{9}{40} = 0.225$$

$$\text{For 5 games, } \frac{7}{40} = 0.175$$

$$\text{For 7 games, } \frac{16}{40} = 0.400$$

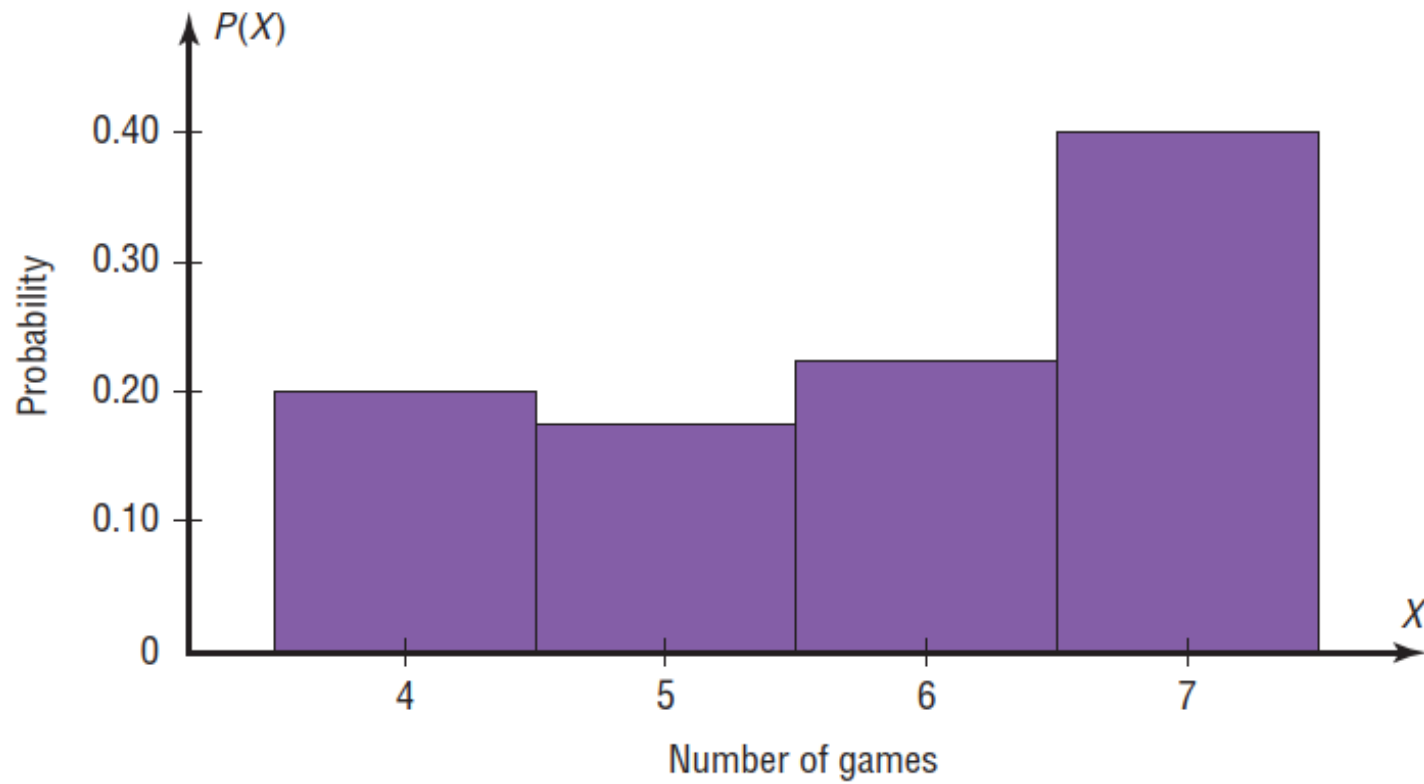
The probability distribution is

Number of games X	4	5	6	7
Probability $P(X)$	0.200	0.175	0.225	0.400



Example 1

SOLUTION



Requirements for a discrete probability distribution

Two Requirements for a Probability Distribution

1. The sum of the probabilities of all the events in the sample space must equal 1; that is, $\sum P(X) = 1$.
2. The probability of each event in the sample space must be between or equal to 0 and 1. That is, $0 \leq P(X) \leq 1$.



Example 2

Determine whether each distribution is a probability distribution.

a.

X	5	8	11	14
$P(X)$	0.2	0.6	0.1	0.3

b.

X	1	2	3	4	5
$P(X)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

c.

X	1	2	3	4
$P(X)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

d.

X	4	8	12
$P(X)$	-0.5	0.6	0.4



Example 2

SOLUTION

- a.* No. The sum of the probabilities is greater than 1.
- b.* Yes. The sum of the probabilities of all the events is equal to 1. Each probability is greater than or equal to 0 and less than or equal to 1.
- c.* Yes. The sum of the probabilities of all the events is equal to 1. Each probability is greater than or equal to 0 and less than or equal to 1.
- d.* No. One of the probabilities is less than 0.



Mean of random variable with a discrete probability distribution

Formula for the Mean of a Probability Distribution

The mean of a random variable with a discrete probability distribution is

$$\begin{aligned}\mu &= X_1 \cdot P(X_1) + X_2 \cdot P(X_2) + X_3 \cdot P(X_3) + \cdots + X_n \cdot P(X_n) \\ &= \sum X \cdot P(X)\end{aligned}$$

where $X_1, X_2, X_3, \dots, X_n$ are the outcomes and $P(X_1), P(X_2), P(X_3), \dots, P(X_n)$ are the corresponding probabilities.

Note: $\sum X \cdot P(X)$ means to sum the products.



Example 3

Find the mean of the number of spots that appear when a die is tossed.

SOLUTION

In the toss of a die, the mean can be computed thus.

Outcome X	1	2	3	4	5	6
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\begin{aligned}\mu &= \sum X \cdot P(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} \\ &= \frac{21}{6} = 3\frac{1}{2} \text{ or } 3.5\end{aligned}$$



Example 4

In families with five children, find the mean number of children who will be girls.

SOLUTION

First, it is necessary to find the probability distribution for the number of females. There are 32 outcomes, and there is one way for a family to have no girls. There are five ways to have one girl, that is, FMMMM, MFMMM, MMFMM, MMMFM, MMMMF. Continue with two females and three males, three females and two males, four females and one male, and five females. You can draw a tree diagram to help you.

The probability distribution is

Number of girls X	0	1	2	3	4	5
Probability $P(X)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

The mean is

$$\mu = \sum X \cdot P(X) = 0 \cdot \frac{1}{32} + 1 \cdot \frac{5}{32} + 2 \cdot \frac{10}{32} + 3 \cdot \frac{10}{32} + 4 \cdot \frac{5}{32} + 5 \cdot \frac{1}{32} = 2\frac{1}{2} = 2.5$$

Hence, the mean number of females is 2.5.



Example 5, revisit slides 5 and 6

If three coins are tossed, find the mean of the number of heads that occur.

SOLUTION

The probability distribution is

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

The mean is

$$\mu = \sum X \cdot P(X) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1\frac{1}{2} \text{ or } 1.5$$

The value 1.5 cannot occur as an outcome. Nevertheless, it is the long-run or theoretical average.



Example 6

The probability distribution shown represents the number of trips of five nights or more that American adults take per year. (That is, 6% do not take any trips lasting five nights or more, 70% take one trip lasting five nights or more per year, etc.) Find the mean.

Number of trips X	0	1	2	3	4
Probability $P(X)$	0.06	0.70	0.20	0.03	0.01

SOLUTION

$$\begin{aligned}\mu &= \sum X \cdot P(X) \\ &= (0)(0.06) + (1)(0.70) + (2)(0.20) + (3)(0.03) + (4)(0.01) \\ &= 0 + 0.70 + 0.40 + 0.09 + 0.04 \\ &= 1.23\end{aligned}$$

Hence, the mean of the number of trips lasting five nights or more per year taken by American adults is 1.23.



Variance of a discrete probability distribution

Formula for the Variance of a Probability Distribution

Find the variance of a probability distribution by multiplying the square of each outcome by its corresponding probability, summing those products, and subtracting the square of the mean. The formula for the variance of a probability distribution is

$$\sigma^2 = \sum[X^2 \cdot P(X)] - \mu^2$$

The standard deviation of a probability distribution is

$$\sigma = \sqrt{\sigma^2} \quad \text{or} \quad \sigma = \sqrt{\sum[X^2 \cdot P(X)] - \mu^2}$$



Example 7

A box contains 5 balls. Two are numbered 3, one is numbered 4, and two are numbered 5. The balls are mixed and one is selected at random. After a ball is selected, its number is recorded. Then it is replaced. If the experiment is repeated many times, find the variance and standard deviation of the numbers on the balls.

SOLUTION

Let X be the number on each ball. The probability distribution is

Number on ball X	3	4	5
Probability $P(X)$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$



Example 7

SOLUTION

The mean is

$$\mu = \sum X \cdot P(X) = 3 \cdot \frac{2}{5} + 4 \cdot \frac{1}{5} + 5 \cdot \frac{2}{5} = 4$$

The variance is

$$\begin{aligned}\sigma^2 &= \sum [X^2 \cdot P(X)] - \mu^2 \\ &= 3^2 \cdot \frac{2}{5} + 4^2 \cdot \frac{1}{5} + 5^2 \cdot \frac{2}{5} - 4^2 \\ &= 16\frac{4}{5} - 16 \\ &= \frac{4}{5} \text{ or } 0.8\end{aligned}$$

The standard deviation is

$$\sigma = \sqrt{\frac{4}{5}} = \sqrt{0.8} \approx 0.894$$



Example 7

SOLUTION The mean, variance, and standard deviation can also be found by using vertical columns, as shown.

X	$P(X)$	$X \cdot P(X)$	$X^2 \cdot P(X)$
3	0.4	1.2	3.6
4	0.2	0.8	3.2
5	0.4	<u>2.0</u>	<u>10</u>
		$\Sigma X \cdot P(X) = 4.0$	16.8

Find the mean by summing the $\Sigma X \cdot P(X)$ column, and find the variance by summing the $X^2 \cdot P(X)$ column and subtracting the square of the mean.

$$\sigma^2 = 16.8 - 4^2 = 16.8 - 16 = 0.8$$

and

$$\sigma = \sqrt{0.8} \approx 0.894$$



Example 8

A talk radio station has four telephone lines. If the host is unable to talk (i.e., during a commercial) or is talking to a person, the other callers are placed on hold. When all lines are in use, others who are trying to call in get a busy signal. The probability that 0, 1, 2, 3, or 4 people will get through is shown in the probability distribution. Find the variance and standard deviation for the distribution.

X	0	1	2	3	4
$P(X)$	0.18	0.34	0.23	0.21	0.04

Should the station have considered getting more phone lines installed?



Example 8

SOLUTION

The mean is

$$\begin{aligned}\mu &= \sum X \cdot P(X) \\ &= 0 \cdot (0.18) + 1 \cdot (0.34) + 2 \cdot (0.23) + 3 \cdot (0.21) + 4 \cdot (0.04) \\ &= 1.59\end{aligned}$$

The variance is

$$\begin{aligned}\sigma^2 &= \sum [X^2 \cdot P(X)] - \mu^2 \\ &= [0^2 \cdot (0.18) + 1^2 \cdot (0.34) + 2^2 \cdot (0.23) + 3^2 \cdot (0.21) + 4^2 \cdot (0.04)] - 1.59^2 \\ &= (0 + 0.34 + 0.92 + 1.89 + 0.64) - 2.528 \\ &= 3.79 - 2.528 = 1.262\end{aligned}$$



Example 8

SOLUTION

The standard deviation is $\sigma = \sqrt{\sigma^2}$, or $\sigma = \sqrt{1.262} \approx 1.123$.

No. The mean number of people calling at any one time is 1.59. Since the standard deviation is 1.123, most callers would be accommodated by having four phone lines because $\mu + 2\sigma$ would be $1.59 + 2(1.123) = 3.836 \approx 4.0$. Very few callers would get a busy signal since at least 75% of the callers would either get through or be put on hold.



Example 9: Expectation

The **expected value** of a discrete random variable of a probability distribution is the theoretical average of the variable. The formula is

$$\mu = E(X) = \sum X \cdot P(X)$$

The symbol $E(X)$ is used for the expected value.

One thousand tickets are sold at \$1 each for a color television valued at \$350. What is the expected value of the gain if you purchase one ticket?



Example 9: Expectation

SOLUTION

The problem can be set up as follows:

	Win	Lose
Gain X	\$349	−\$1
Probability $P(X)$	$\frac{1}{1000}$	$\frac{999}{1000}$

Two things should be noted. First, for a win, the net gain is \$349, since you do not get the cost of the ticket (\$1) back. Second, for a loss, the gain is represented by a negative number, in this case −\$1. The solution, then, is

$$E(X) = \$349 \cdot \frac{1}{1000} + (-\$1) \cdot \frac{999}{1000} = -\$0.65$$

Hence, a person would lose, on average, \$0.65 on each ticket purchased.



Example 10: Expectation

Six balls numbered 1, 2, 3, 5, 8, and 13 are placed in a box. A ball is selected at random, and its number is recorded and then it is replaced. Find the expected value of the numbers that will occur.

SOLUTION

Since the balls are replaced, the probability for each number is $\frac{1}{6}$, so the probability distribution is

Number (X)	1	2	3	5	8	13
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

The expected value is

$$E(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 8 \cdot \frac{1}{6} + 13 \cdot \frac{1}{6} = 5\frac{1}{3}$$



The Binomial, Poisson, and Geometric Distributions



The binomial experiment

A **binomial experiment** is a probability experiment that satisfies the following four requirements:

1. There must be a fixed number of trials.
2. Each trial can have only two outcomes or outcomes that can be reduced to two outcomes. These outcomes can be considered as either success or failure.
3. The outcomes of each trial must be independent of one another.
4. The probability of a success must remain the same for each trial.



Example 11: The binomial experiment

Decide whether each experiment is a binomial experiment. If not, state the reason why.

- a.* Selecting 20 university students and recording their class rank
- b.* Selecting 20 students from a university and recording their gender
- c.* Drawing five cards from a deck without replacement and recording whether they are red or black cards
- d.* Selecting five students from a large school and asking them if they are on the dean's list
- e.* Recording the number of children in 50 randomly selected families



Example 11: The binomial experiment

SOLUTION

- a.* No. There are five possible outcomes: freshman, sophomore, junior, senior, and graduate student.
- b.* Yes. All four requirements are met.
- c.* No. Since the cards are not replaced, the events are not independent.
- d.* Yes. All four requirements are met.
- e.* No. There can be more than two categories for the answers.



The binomial distribution

The outcomes of a binomial experiment and the corresponding probabilities of these outcomes are called a **binomial distribution**.

Notation for the Binomial Distribution

$P(S)$ The symbol for the probability of success

$P(F)$ The symbol for the probability of failure

p The numerical probability of a success

q The numerical probability of a failure

$$P(S) = p \quad \text{and} \quad P(F) = 1 - p = q$$

n The number of trials

X The number of successes in n trials

Page - 32 Note that $0 \leq X \leq n$ and $X = 0, 1, 2, 3, \dots, n$.



The binomial distribution

Binomial Probability Formula

In a binomial experiment, the probability of exactly X successes in n trials is

$$P(X) = \frac{n!}{(n - X)!X!} \cdot p^X \cdot q^{n-X}$$

In general, the number of ways to get X successes from n trials without regard to order is

$${}_nC_X = \frac{n!}{(n - X)!X!}$$



Example 12: The binomial distribution

A coin is tossed 3 times. Find the probability of getting exactly two heads.

SOLUTION

This problem can be solved by looking at the sample space. There are three ways to get two heads.

HHH, HHT, HTH, THH, TTH, THT, HTT, TTT

The answer is $\frac{3}{8}$, or 0.375.



Example 12: The binomial distribution

SOLUTION

Looking at the problem in Example 12 from the standpoint of a binomial experiment, one can show that it meets the four requirements.

1. There are a fixed number of trials (three).
2. There are only two outcomes for each trial, heads or tails.
3. The outcomes are independent of one another (the outcome of one toss in no way affects the outcome of another toss).
4. The probability of a success (heads) is $\frac{1}{2}$ in each case.

In this case, $n = 3$, $X = 2$, $p = \frac{1}{2}$, and $q = \frac{1}{2}$. Hence, substituting in the formula gives

$$P(2 \text{ heads}) = \frac{3!}{(3 - 2)!2!} \cdot \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = \frac{3}{8} = 0.375$$

Page - 35 which is the same answer obtained by using the sample space.



Example 13: The binomial distribution

A survey found that one out of five Americans says he or she has visited a doctor in any given month. If 10 people are selected at random, find the probability that exactly 3 will have visited a doctor last month.

SOLUTION

In this case, $n = 10$, $X = 3$, $p = \frac{1}{5}$, and $q = \frac{4}{5}$. Hence,

$$P(3) = \frac{10!}{(10 - 3)!3!} \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^7 \approx 0.201$$

So, there is a 0.201 probability that in a random sample of 10 people, exactly 3 of them visited a doctor in the last month.



Example 14: The binomial distribution

A survey from Teenage Research Unlimited (Northbrook, Illinois) found that 30% of teenage consumers receive their spending money from part-time jobs. If 5 teenagers are selected at random, find the probability that at least 3 of them will have part-time jobs.

SOLUTION

To find the probability that at least 3 have part-time jobs, it is necessary to find the individual probabilities for 3, or 4, or 5 and then add them to get the total probability.

$$P(3) = \frac{5!}{(5-3)!3!} (0.3)^3(0.7)^2 \approx 0.132$$

$$P(4) = \frac{5!}{(5-4)!4!} (0.3)^4(0.7)^1 \approx 0.028$$

$$P(5) = \frac{5!}{(5-5)!5!} (0.3)^5(0.7)^0 \approx 0.002$$

Hence,

$$\begin{aligned} P(\text{at least three teenagers have part-time jobs}) \\ = 0.132 + 0.028 + 0.002 = 0.162 \end{aligned}$$



Example 15: The binomial distribution

Public Opinion reported that 5% of Americans are afraid of being alone in a house at night. If a random sample of 20 Americans is selected, find these probabilities.

- a. There are exactly 5 people in the sample who are afraid of being alone at night.
- b. There are at most 3 people in the sample who are afraid of being alone at night.
- c. There are at least 3 people in the sample who are afraid of being alone at night.



Example 15: The binomial distribution

SOLUTION

- a. $n = 20$, $p = 0.05$, and $X = 5$, we get 0.002.
- b. $n = 20$ and $p = 0.05$. “At most 3 people” means 0, or 1, or 2, or 3.

Hence, the solution is

$$\begin{aligned}P(X \leq 3) &= P(0) + P(1) + P(2) + P(3) = 0.358 + 0.377 + 0.189 + 0.060 \\&= 0.984\end{aligned}$$

- c. $n = 20$ and $p = 0.05$. “At least 3 people” means 3, 4, 5, . . . , 20. This problem can best be solved by finding $P(X \leq 2) = P(0) + P(1) + P(2)$ and subtracting from 1.

$$P(0) + P(1) + P(2) = 0.358 + 0.377 + 0.189 = 0.924$$

$$P(X \geq 3) = 1 - 0.924 = 0.076$$



Example 16: Mean, variance, and standard deviation of binomial distributions

Mean, Variance, and Standard Deviation for the Binomial Distribution

The mean, variance, and standard deviation of a variable that has the *binomial distribution* can be found by using the following formulas.

$$\text{Mean: } \mu = n \cdot p \quad \text{Variance: } \sigma^2 = n \cdot p \cdot q \quad \text{Standard deviation: } \sigma = \sqrt{n \cdot p \cdot q}$$

A coin is tossed 4 times. Find the mean, variance, and standard deviation of the number of heads that will be obtained.

SOLUTION

With the formulas for the binomial distribution and $n = 4$, $p = \frac{1}{2}$, and $q = \frac{1}{2}$, the results are

$$\mu = n \cdot p = 4 \cdot \frac{1}{2} = 2$$

$$\sigma^2 = n \cdot p \cdot q = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$$

$$\sigma = \sqrt{1} = 1$$

In this case, the mean is two heads. The variance is 1 and the standard deviation is 1.

Example 16: Mean, variance, and standard deviation of binomial distributions

A coin is tossed 4 times. Find the mean, variance, and standard deviation of the number of heads that will be obtained.

SOLUTION

As stated previously, this problem can be solved by using the formulas for expected value. The distribution is shown.

No. of heads X	0	1	2	3	4
Probability $P(X)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$$\mu = E(X) = \sum X \cdot P(X) = 0 \cdot \frac{1}{16} + 1 \cdot \frac{4}{16} + 2 \cdot \frac{6}{16} + 3 \cdot \frac{4}{16} + 4 \cdot \frac{1}{16} = \frac{32}{16} = 2$$

$$\sigma^2 = \sum X^2 \cdot P(X) - \mu^2$$

$$= 0^2 \cdot \frac{1}{16} + 1^2 \cdot \frac{4}{16} + 2^2 \cdot \frac{6}{16} + 3^2 \cdot \frac{4}{16} + 4^2 \cdot \frac{1}{16} - 2^2 = \frac{80}{16} - 4 = 1$$

$$\sigma = \sqrt{1} = 1$$

Hence, the simplified binomial formulas give the same results.



Example 17: Mean, variance, and standard deviation of binomial distributions

An 8-sided die (with the numbers 1 through 8 on the faces) is rolled 560 times. Find the mean, variance, and standard deviation of the number of 7s that will be rolled.

SOLUTION

This is a binomial experiment with $n = 560$, $p = \frac{1}{8}$, and $q = \frac{7}{8}$ so that

$$\mu = n \cdot p = 560 \cdot \frac{1}{8} = 70$$

$$\sigma^2 = n \cdot p \cdot q = 560 \cdot \frac{1}{8} \cdot \frac{7}{8} = 61\frac{1}{4} = 61.25$$

$$\sigma = \sqrt{61.25} = 7.826$$

In this case, the mean of the number of 7s obtained is 70. The variance is 61.25, and the standard deviation is 7.826.



Example 18: Mean, variance, and standard deviation of binomial distributions

The *Statistical Bulletin* published by Metropolitan Life Insurance Co. reported that 2% of all American births result in twins. If a random sample of 8000 births is taken, find the mean, variance, and standard deviation of the number of births that would result in twins.

SOLUTION

This is a binomial situation, since a birth can result in either twins or not twins (i.e., two outcomes).

$$\mu = n \cdot p = (8000)(0.02) = 160$$

$$\sigma^2 = n \cdot p \cdot q = (8000)(0.02)(0.98) = 156.8$$

$$\sigma = \sqrt{n \cdot p \cdot q} = \sqrt{156.8} = 12.522$$

For the sample, the average number of births that would result in twins is 160; the variance is 156.8, or 157; and the standard deviation is 12.522, or 13 if rounded.



The Poisson experiment

A **Poisson experiment** is a probability experiment that satisfies the following requirements:

1. The random variable X is the number of occurrences of an event over some interval (i.e., length, area, volume, period of time, etc.).
2. The occurrences occur randomly.
3. The occurrences are independent of one another.
4. The average number of occurrences over an interval is known.



The Poisson distribution

Formula for the Poisson Distribution

The probability of X occurrences in an interval of time, volume, area, etc., for a variable where λ (Greek letter lambda) is the mean number of occurrences per unit (time, volume, area, etc.) is

$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} \quad \text{where } X = 0, 1, 2, \dots$$

The letter e is a constant approximately equal to 2.7183.



Portions of the Poisson distribution table

The Poisson Distribution

x	λ									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	.9048	.8187	.7408	.6703	.6065	.5488	.4966	.4493	.4066	.3679
1	.0905	.1637	.2222	.2681	.3033	.3293	.3476	.3595	.3659	.3679
2	.0045	.0164	.0333	.0536	.0758	.0988	.1217	.1438	.1647	.1839
3	.0002	.0011	.0033	.0072	.0126	.0198	.0284	.0383	.0494	.0613
4	.0000	.0001	.0003	.0007	.0016	.0030	.0050	.0077	.0111	.0153
5	.0000	.0000	.0000	.0001	.0002	.0004	.0007	.0012	.0020	.0031
6	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0002	.0003	.0005
7	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001



Portions of the Poisson distribution table

x	λ									
	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	.3329	.3012	.2725	.2466	.2231	.2019	.1827	.1653	.1496	.1353
1	.3662	.3614	.3543	.3452	.3347	.3230	.3106	.2975	.2842	.2707
2	.2014	.2169	.2303	.2417	.2510	.2584	.2640	.2678	.2700	.2707
3	.0738	.0867	.0998	.1128	.1255	.1378	.1496	.1607	.1710	.1804
4	.0203	.0260	.0324	.0395	.0471	.0551	.0636	.0723	.0812	.0902
5	.0045	.0062	.0084	.0111	.0141	.0176	.0216	.0260	.0309	.0361
6	.0008	.0012	.0018	.0026	.0035	.0047	.0061	.0078	.0098	.0120
7	.0001	.0002	.0003	.0005	.0008	.0011	.0015	.0020	.0027	.0034
8	.0000	.0000	.0001	.0001	.0001	.0002	.0003	.0005	.0006	.0009
9	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0001	.0001	.0002



Portions of the Poisson distribution table

x	λ									
	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
0	.1225	.1108	.1003	.0907	.0821	.0743	.0672	.0608	.0550	.0498
1	.2572	.2438	.2306	.2177	.2052	.1931	.1815	.1703	.1596	.1494
2	.2700	.2681	.2652	.2613	.2565	.2510	.2450	.2384	.2314	.2240
3	.1890	.1966	.2033	.2090	.2138	.2176	.2205	.2225	.2237	.2240
4	.0992	.1082	.1169	.1254	.1336	.1414	.1488	.1557	.1622	.1680
5	.0417	.0476	.0538	.0602	.0668	.0735	.0804	.0872	.0940	.1008
6	.0146	.0174	.0206	.0241	.0278	.0319	.0362	.0407	.0455	.0504
7	.0044	.0055	.0068	.0083	.0099	.0118	.0139	.0163	.0188	.0216
8	.0011	.0015	.0019	.0025	.0031	.0038	.0047	.0057	.0068	.0081
9	.0003	.0004	.0005	.0007	.0009	.0011	.0014	.0018	.0022	.0027
10	.0001	.0001	.0001	.0002	.0002	.0003	.0004	.0005	.0006	.0008
11	.0000	.0000	.0000	.0000	.0000	.0001	.0001	.0001	.0002	.0002
12	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0001



Portions of the Poisson distribution table

x	λ									
	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4.0
0	.0450	.0408	.0369	.0334	.0302	.0273	.0247	.0224	.0202	.0183
1	.1397	.1304	.1217	.1135	.1057	.0984	.0915	.0850	.0789	.0733
2	.2165	.2087	.2008	.1929	.1850	.1771	.1692	.1615	.1539	.1465
3	.2237	.2226	.2209	.2186	.2158	.2125	.2087	.2046	.2001	.1954
4	.1734	.1781	.1823	.1858	.1888	.1912	.1931	.1944	.1951	.1954



Example 19: The Poisson distribution

If there are 200 typographical errors randomly distributed in a 500-page manuscript, find the probability that a given page contains exactly 3 errors.

SOLUTION

First, find the mean number λ of errors. Since there are 200 errors distributed over 500 pages, each page has an average of

$$\lambda = \frac{200}{500} = \frac{2}{5} = 0.4$$

or 0.4 error per page. Since $X = 3$, substituting into the formula yields

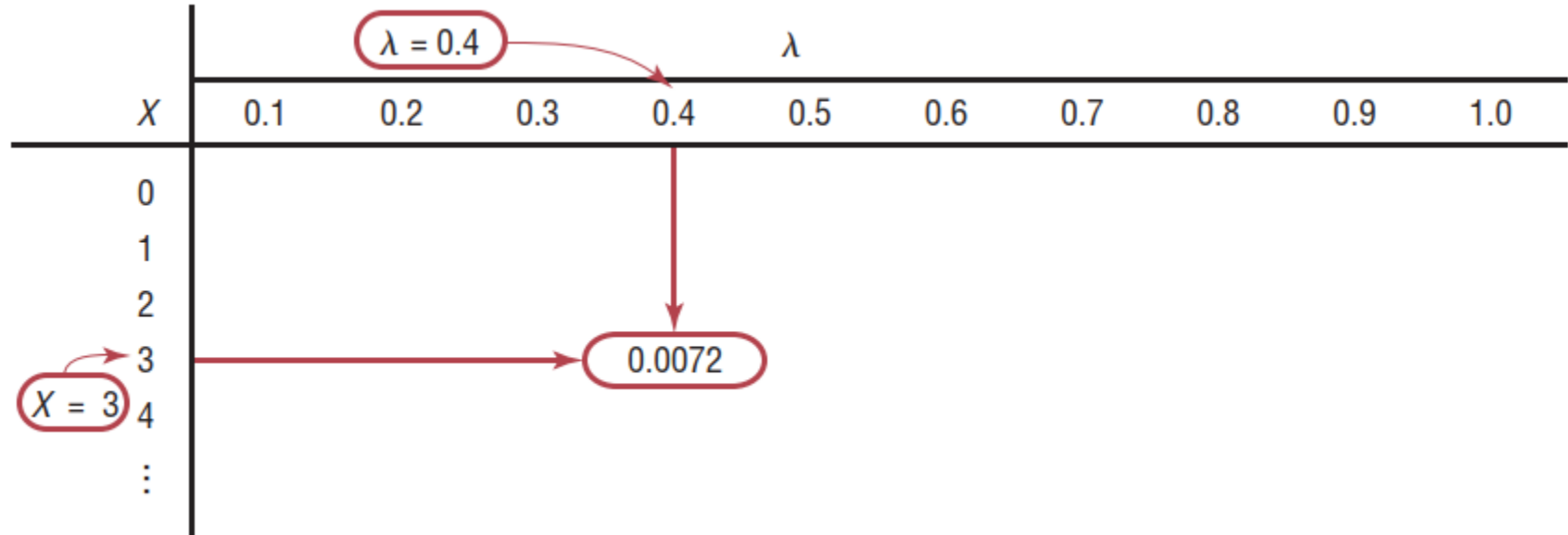
$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} = \frac{(2.7183)^{-0.4} (0.4)^3}{3!} \approx 0.0072$$

Thus, there is less than a 1% chance that any given page will contain exactly 3 errors.



Example 19: The Poisson distribution

SOLUTION



Example 20: The Poisson distribution, revisit slide 48

A sales firm receives, on average, 3 calls per hour on its toll-free number. For any given hour, find the probability that it will receive the following.

a. At most 3 calls

b. At least 3 calls

c. 5 or more calls

SOLUTION

a. “At most 3 calls” means 0, 1, 2, or 3 calls. Hence,

$$\begin{aligned} &P(0; 3) + P(1; 3) + P(2; 3) + P(3; 3) \\ &= 0.0498 + 0.1494 + 0.2240 + 0.2240 \\ &= 0.6472 \end{aligned}$$



Example 20: The Poisson distribution, revisit slide 48

SOLUTION

- b. “At least 3 calls” means 3 or more calls. It is easier to find the probability of 0, 1, and 2 calls and then subtract this answer from 1 to get the probability of at least 3 calls.

$$P(0; 3) + P(1; 3) + P(2; 3) = 0.0498 + 0.1494 + 0.2240 = 0.4232$$

and

$$1 - 0.4232 = 0.5768$$

- c. For the probability of 5 or more calls, it is easier to find the probability of getting 0, 1, 2, 3, or 4 calls and subtract this answer from 1. Hence,

$$\begin{aligned} P(0; 3) + P(1; 3) + P(2; 3) + P(3; 3) + P(4; 3) \\ = 0.0498 + 0.1494 + 0.2240 + 0.2240 + 0.1680 \\ = 0.8152 \end{aligned}$$

and

$$1 - 0.8152 = 0.1848$$

Thus, for the events described, the part *a* event is most likely to occur, and the part *c* event is least likely to occur.



The geometric distribution

A **geometric experiment** is a probability experiment if it satisfies the following requirements:

1. Each trial has two outcomes that can be either success or failure.
2. The outcomes are independent of each other.
3. The probability of a success is the same for each trial.
4. The experiment continues until a success is obtained.

Formula for the Geometric Distribution

If p is the probability of a success on each trial of a binomial experiment and n is the number of the trial at which the first success occurs, then the probability of getting the first success on the n th trial is

$$P(n) = p(1 - p)^{n-1}$$

where $n = 1, 2, 3, \dots$



Example 21: The geometric distribution

A coin is tossed. Find the probability of getting the first head on the third toss.

SOLUTION

The objective for tossing a coin and getting a head on the third toss is TTH. The probability for this outcome is

$$\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$$

Now by using the formula, you get the same results.

$$\begin{aligned} P(n) &= p(1 - p)^{n-1} = \frac{1}{2}\left(1 - \frac{1}{2}\right)^{3-1} \\ &= \frac{1}{2}\left(\frac{1}{2}\right)^2 = \frac{1}{8} \end{aligned}$$

Hence, there is a 1 out of 8 chance or 0.125 probability of getting the first head on the third toss of a coin.



Example 22: The geometric distribution

In the United States, approximately 42% of people have type A blood. If 4 people are selected at random, find the probability that the fourth person is the first one selected with type A blood.

SOLUTION

Let $p = 0.42$ and $n = 4$.

$$P(n) = p(1 - p)^{n-1}$$

$$\begin{aligned} P(4) &= (0.42)(1 - 0.42)^{4-1} = (0.42)(0.58)^3 \\ &\approx 0.0819 \approx 0.082 \end{aligned}$$

There is a 0.082 probability that the fourth person selected will be the first one to have type A blood.



Summary of discrete distributions

Summary of Discrete Distributions

1. Binomial distribution

$$P(X) = \frac{n!}{(n-X)!X!} \cdot p^X \cdot q^{n-X}$$
$$\mu = n \cdot p \quad \sigma = \sqrt{n \cdot p \cdot q}$$

It is used when there are only two outcomes for a fixed number of independent trials and the probability for each success remains the same for each trial.

2. Poisson distribution

$$P(X; \lambda) = \frac{e^{-\lambda} \lambda^X}{X!} \quad \text{where } X = 0, 1, 2, \dots$$

It is used when n is large and p is small, and the independent variable occurs over a period of time, or a density of items is distributed over a given area or volume.

3. Geometric distribution

$$P(n) = p(1-p)^{n-1} \quad \text{where } n = 1, 2, 3, \dots$$

It is used when there are two outcomes and we are interested in the probability that the first success occurs on the n th trial.

